



Force is a very common concept used in Engineering Mechanics. There are different types and classifications of forces.

Gravitational Force

A force that is exerted by the earth's gravity on a certain object. Normally, this force is represented by the weight of the object, a force that is directed towards the center of the earth.

The weight of the object is defined as:

$$W = mg \quad g = 9.81 \text{ m/s}^2 \\ g = 32.2 \text{ ft/s}^2$$

Normal Force

A force that is perpendicular to the surface of contact, if there are two objects or planes that are in contact with each other.

Excel Review Center

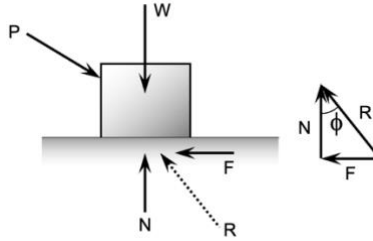
Friction Force

A force that is parallel to the surface of contact, if there are two objects or planes that are in contact with each other. It is also defined as the limited amount of resistance to sliding between the surfaces of two bodies in contact. Friction is oftentimes defined as the force that tends to oppose the motion, hence its direction is opposite the direction of the actual motion (for **kinetic friction**) or the impending motion (for **static friction**). Kinetic friction is the friction against a moving object while static friction is the friction against a stationary object.

The **coefficient of static friction** depends upon the type of materials or the nature of surfaces in contact. It is independent of normal reaction, length and area of contact.

$$F = \mu N \quad \tan \phi = \mu$$

Excel Review Center



where: N = normal force
F = frictional force
R = resultant force
 μ = coefficient of friction
 ϕ = angle of friction

Dynamic friction is the friction for a body in motion. It is always less than the static friction.

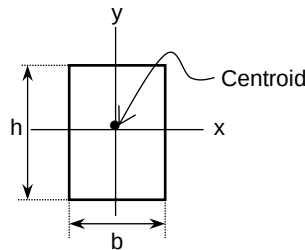
Moment of Inertia

Moment of inertia is also called the **second moment of area**.

With Respect To Centroid Axis:

Centroidal moment of inertia is the moment of inertia of the figure with respect to an axis passing through the centroid.

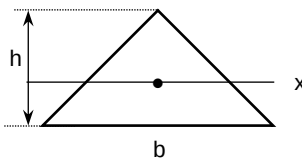
A. Rectangle



$$I_x = \frac{bh^3}{12}$$

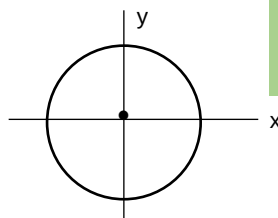
$$I_y = \frac{b^3h}{12}$$

B. Triangle



$$I_x = \frac{bh^3}{36}$$

C. Circle

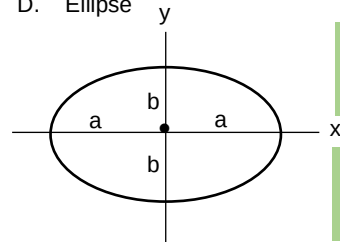


$$I_x = \frac{\pi r^4}{4} = \frac{\pi D^4}{64}$$

$$I_x = I_y$$

where: r = radius of circle
D = diameter of circle

D. Ellipse



$$I_x = \frac{\pi ab^3}{4}$$

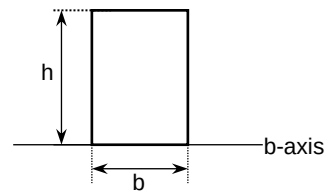
$$I_y = \frac{\pi a^3b}{4}$$

where: a = length of semi-major axis
b = length of semi-minor axis

With Respect To The Base

A. Rectangle

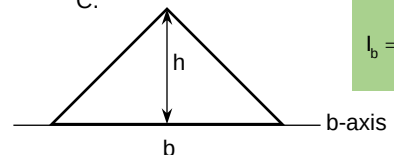
Excel Review



$$I_b = \frac{bh^3}{3}$$

B. Triangle

C.



$$I_b = \frac{bh^3}{12}$$

For composite figures and for axis not at the centroid nor at the base, moment of inertia may be calculated using the **transfer formula**, which is as follows;

$$I = I_g + Ad^2$$

where: d = distance from the centroid to the axis
A = area of the figure

